scientific reports

OPEN

The associations of social isolation with depression and anxiety among adults aged 65 years and older in Ningbo, China

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Social isolation was associated with emotional problems (depression and anxiety) among older adults, however, little is known in China. Thus, we conducted a cross-sectional study including 6,664 ≥ 65 years older adults in Ningbo, China. We collected data on social isolation, depression, and anxiety by specific scales. The relationship between social isolation and emotional problems was estimated by multivariate-adjusted logistic regression models. The population–attributable risk percentage (PAR%) was used to explore the contribution of social isolation to emotional problems. Overall, the percentage of participants who had experienced social isolation, depression, and anxiety was 12.67%, 4.83%, and 2.63%. Compared with the elderly without social isolation, the adjusted odds ratios (95% confidence interval) of depression and anxiety with social isolation were 1.77 (1.25–2.51) and 1.66 (1.05–2.63), respectively. The PAR analysis showed that 10.66% of depression and 9.03% of anxiety could be attributable to social isolation. In the gender subgroup, ORs and PAR% were only significantly observed in female participants. In Chinese older adults, social isolation has been linked to depression and anxiety, suggesting the importance of taking effective and feasible interventions to reduce social isolation and emotional problems, especially among females.

Depression and anxiety are the most frequent emotional problems experienced by older adults worldwide[1]. In China, nearly 3.8% and 4.7% of older adults (over 65 years) were reported to suffer from depressive and anxiety disorders, respectively[2], resulting in a high burden of disease in this population. Depression and anxiety are generated both by biological factors (e.g., sex, somatic illness, functional disability) and psychosocial factors (e.g., low contact frequency, a childless or unmarried status)[3]. Changes in some psychosocial factors may reduce emotional problems in older adults[3].

Social isolation refers to a poor living condition involving a lack of social contact and participation or inadequate social interactions with friends, relatives, and others[4]. Social isolation can be measured by the following variables: whether one lives alone, the frequency of conversations with relatives and friends, and the frequency of participation in social activities[5]. A study conducted in the United States found that approximately 24.0% of community-dwelling adults older than 65 years old were socially isolated[6]. Two studies conducted in China showed that the prevalence of social isolation was 29.7% among community-dwelling older adults in Qingdao[7] and 28.9% among their counterparts in Shanghai[8].

Studies generally have recognized the profound negative effect of social isolation on physical and mental health[9,10]. The longitudinal Aging Social Survey found that social isolation independently determined the probability of depression in both China[11] and the Netherlands[12]. A study in Ireland[13] found that objective social isolation independently affected the process of depression or anxiety. However, no study has explored the proportion of cases of depression and anxiety that might be prevented by alleviating social isolation among older adults. Previous research has indicated that various factors are linked to depression and anxiety among older individuals. Age has been consistently identified as a primary contributing factor to the heightened risk of mental health[14,15]. Additionally, social and demographic variables, including education status[15], marital status[16], and household

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income[17] have been shown to correlate with increased prevalence of mental health concerns in older adults. Furthermore, substance use[18], lifestyle behaviors[18,19], and sleep behaviors[20] have been found to have significant associations with mental health outcomes in this population. Self-rated health[21] and chronic disease[16] have also been identified as relevant factors linked to mental health.

Thus, in our study, we aimed to determine the current prevalence of social isolation, depression, and anxiety symptoms among community-dwelling older adults in Ningbo, China, and to investigate the relationships between these variables, including (1) the adjusted associations of social isolation with depression and anxiety, (2) the adjusted population attributable risk percentage (PAR%) for social isolation among cases of depression and anxiety, (3) the adjusted associations and PAR% in the gender subgroup.

Methods

Study design and population

Ningbo, situated in the Yangtze River Delta region of China, is a prominent city with six districts and four counties. As of 2022, its population reached approximately 9.6 million, with 1.4 million individuals aged 65 and older. A cross-sectional study of older adults (aged 65 years and older) in Ningbo, China was conducted from June to August 2022. Two community units in each district and county in Ningbo were randomly selected by computer generation and included all participants in each unit who met the inclusion criteria in our study. The inclusion criteria were (1) residence in the selected communities for more than one year, (2) an age of 65 years or older, and (3) the ability to understand our questionnaires, with no serious mental illness. The exclusion criteria were the absence of key variables in participants. All of the questionnaires were completed by the participants during face-to-face sessions with community workers. For participants who could not complete the paper questionnaire without assistance, the community workers verbally read the questions and completed the questionnaires according to the participants' answers. Ethical approval for this study was obtained from the Ethics Committee of Ningbo Kangning Hospital, and written informed consent was obtained from all participants.

Defining depression and anxiety

Depression and anxiety were assessed using a validated 9-item Patient Health Questionnaire (PHQ-9) and a 7-item validated Generalized Anxiety Disorder (GAD-7) scale; these measures were based on the participants' experiences over the previous 2 weeks, and each item was rated on a 4-point scale: "not at all" = 0, "several days" = 1, "more than half of the days" = 2, and "nearly every day" = 3. The total scores of the PHQ-9 and GAD-7 ranged from 0 to 27 and from 0 to 21, respectively. The Chinese versions of the PHQ-9 and GAD-7 have both been shown to have good reliability and validity for screening anxiety[22] and depression[23]. As previously recommended[24,25], we used the screening cutoff score of 10 for both scales, and a score ≥ 10 corresponded to at least a moderate level of depression or anxiety. A PHQ-9 score 10 had a sensitivity (the probability of a positive test given that the elderly has the disease) of 88% and a specificity (the probability of a negative test given that the elderly is well) of 88% for major depression[26], A GAD-7 score 10 had a sensitivity of 89% and a specificity of 82% for generalized anxiety disorder[25].

Defining social isolation

Social isolation was assessed using three questions that addressed whether the participant lived alone (yes = 1, no = 0), had social contact (contacted less than once per month = 1, other = 0), and participated in social activities (participated less than once per week = 1, other = 0); these questions have been described elsewhere[5,27]. We then summed the scores of the responses to these questions, and a score of 2 or 3 was considered to indicate social isolation[27].

Covariates

All of the covariates were based on baseline data. The sociodemographic characteristics included age (continuous), gender (male, female), residence location (urban, rural), education (< 6 or ≥ 6 years), marital status (married or widowed/divorced/never married), and pension income (< 2000, 2000–5000, or > 5000 CNY/month). Additional covariates included the participants' self-rated health (good, intermediate, or poor) and number of chronic non-communicable diseases (0, 1, 2, ≥ 3).

Substance use included two aspects: alcohol consumption (never, former, or current) and smoking (never, former, or current). Lifestyle behaviors included three aspects: sedentary time per day (< 3, 3–5, or > 5 h/day), physical activity performance per week (< 3 or ≥ 3 times/week), and a self-reported healthy diet (yes or no). Sleep behaviors included two aspects: sleep disturbance (difficulty falling asleep or waking up after falling asleep) over the past week (0, 1–2, 3–4, or 5–7 times/week) and sleep duration over the past week (< 6, 6–8, > 8 h/day).

Statistical analysis

The participants' baseline characteristics were compared using the *t*-test for continuous variables and the chi-squared test for categorical variables. We estimated the proportions of participants with social isolation, depression, and anxiety in all population and gender subgroup. The links between social isolation and depression and anxiety were computed using a multivariate-adjusted logistic regression model and are reported as adjusted odds ratios (AORs). Population attributable risk percentages (PAR%)[28] were calculated to assess the proportions of cases of depression and anxiety that could potentially be prevented if social isolation were mitigated among older adults. The calculation was based on the formula reported by Bruzzi et al.[28]: $PAR\% = 1 - \sum_j \frac{P_j}{\widetilde{R_j}}$, where *j* is 1, 2,

3, … level; P_j is the ratio of the number of cases to the total number of cases at the j level; and $\widetilde{R}_j$ is the relative risk of the exposure factor level adjusted for other factors at the j level. In the statistical analysis, P values < 0.05 were considered statistically significant. All the statistical analyses were completed by SAS version 9.4 (SAS Institute, Cary, NC, USA) .

Ethics statement

This work was conducted in accordance with the principles of the Declaration of Helsinki and ethics approval (NBKNYY-2023-LC-29) was obtained from the Ethics Committee of Ningbo Kangning Hospital, and written informed consent was obtained from all participants.

Results

In total, 6,664 community-dwelling older adults were recruited for this study (response rate was 84.9%), of whom 45.50% were men. In Table 1, the men were significantly older than the women (72.89 ± 6.25 vs. 72.40 ± 6.05 years, $P = 0.001$). The percentages of participants who were current alcohol drinkers and smokers were higher among men than among women (Table 1). The percentage of participants who participated in physical activity more than 3 times/week was lower among men than among women (Table 1). Compared with women, men had better sleep disturbance and sleep duration status ($P < 0.001$ for both) (Table 1).

Overall, 12.67% of the participants reported that they had experienced social isolation, and the rate of social isolation was significantly higher among women than among men (15.66% vs. 9.10%, $P < 0.05$, Fig. 1). The prevalence of depression and anxiety among the participants was 4.83% and 2.63%, respectively. Univariate logistic regression analysis showed that age, sedentary, sleep disturbance, social isolation, and more diseases were significantly positively linked with depression and anxiety. Meanwhile, higher pension income, being in marital status, physical activity, longer sleep duration, healthy diet, and self-rated health were significantly negatively linked with depression and anxiety (Fig. 2).

Multivariate logistic regression analysis showed that older adults who had experienced social isolation were 1.77 times more likely to have depression than those who were not socially isolated (AOR = 1.77, 95% confidence interval [CI]: 1.25–2.51, $P < 0.001$, Table 2). Social isolation also was significantly associated with an increased likelihood of anxiety in older adults (AOR = 1.66, 95% CI: 1.05–2.63, $P = 0.029$, Table 2) after controlling for multiple variables. However, no significant associations of social isolation with depression and anxiety were observed in male participants [AOR (95% CI) = 1.42 (0.77–2.58) and 0.98 (0.42–2.26), respectively; $P > 0.05$ for both]. In contrast, female participants who had experienced social isolation were 1.99 and 1.92 times more likely to have depression and anxiety, respectively, than those who had not experienced social isolation, and these results were significant ($P < 0.05$ for both, Table 2).

The results regarding the adjusted PAR% for social isolation are shown in Table 3. Among our participants, the PAR% related to social isolation was as high as 10.66% and 9.03% for depression and anxiety, respectively. Besides, the adjusted PAR% related to social isolation were as high as 15.12% and 12.84% for depression and anxiety in women. The adjusted PAR% of social isolation were no statistical significance for depression and anxiety in men (Table 4).

Discussion

This study investigated the prevalence of social isolation, depression, and anxiety in a sample of community-dwelling older adults in Ningbo, China. Overall, 12.67% of the participating older adults reported having experienced social isolation, and the prevalence of depression and anxiety in this population was 4.83% and 2.63%, respectively. The participants who had experienced social isolation were more likely to have depression and anxiety than those who were not socially isolated. Furthermore, our results shed light on some key intermediate factors that affected the relationship between social isolation and increased risks of depression and anxiety.

The participants who had experienced social isolation had 1.77 and 1.66 times greater odds of having depression and anxiety than those who were not socially isolated, which was congruent with the findings of earlier studies. For example, in a study of Irish older adults, the lowest level of social isolation was associated with a 43% and 52% probability of depression or anxiety[13]. Furthermore, social disconnectedness could predict an increase in subsequent depression and anxiety symptoms due to an increase in perceived isolation among older adults in the United States[29]. Moreover, social isolation at baseline predicted higher depression and anxiety scores at follow-up (incidence rate ratio = 1.35 and 1.32, respectively) in a study of older adults in Shanghai[8]. As we have seen, it does seem depressive symptoms are understudied in the elderly in social isolation studies in China, but this does not seem to be true for other East Asia generally. In the results of their studies, social isolation has been actively linked to depressive symptoms in the elderly in Japan[30,31] and South Korea[32–34]. Depression and anxiety share a biological mechanism of activation via the hypothalamic–pituitary–adrenal (HPA) axis, which may explain their associations with social isolation[35]. The HPA axis is a major stress response system in humans. Studies have suggested that social isolation increases the activation of the HPA axis in humans[36]. Research findings also have confirmed the possibility of excess HPA axis activity in people with both depression and anxiety [37]. Thus, social isolation may make older adults more vulnerable to depression and anxiety via activation of the HPA axis[35].

In our study, we first found that the PAR% of social isolation was 10.66% and 9.03% for depression and anxiety, respectively, suggesting that eliminating social isolation could reduce depression and anxiety in older adults by almost 10%. In this regard, sufficient attention should be paid to social isolation among older adults, who should be motivated to change their situation and seek relevant help. Moreover, measures to reduce social isolation are needed. Potential interventions for eliminating social isolation among older adults include (1) improving their interpersonal communication skills through lectures and training, (2) increasing regular contact with or the

	Men (n = 3023)	Women (3641)	t/x^2	*P* value
Age	72.89 ± 6.25	72.40 ± 6.05	3.25	0.001
Urban and rural				
Urban	1006 (33.28)	1282 (35.21)	2.73	0.098
Rural	2017 (66.72)	2359 (64.79)		
Education				
< 6 years	1659 (54.88)	1860 (51.08)	9.54	0.002
≥ 6 years	1364 (45.12)	1781 (48.92)		
Marital status				
Married	2730 (90.31)	2844 (78.11)	179.60	< 0.001
Widowed/Divorced/Never married	293 (9.69)	797 (21.89)		
Pension income				
< 2000 CNY/month	1924 (63.69)	2473 (67.95)	141.31	< 0.001
2000–5000 CNY/month	821 (27.18)	1077 (29.59)		
> 5000 CNY/month	276 (9.14)	90 (2.47)		
Self-rated health				
Good	1305 (43.17)	1466 (40.28)	10.21	0.037
Intermediate	1525 (50.45)	1892 (51.99)		
Poor	193 (6.38)	281 (7.72)		
Substance use				
Drinking				
Never	1418 (45.92)	3324 (91.340)	1588.44	< 0.001
Former	537 (17.77)	103 (2.83)		
Current	1067 (35.31))	212 (5.83)		
Smoking				
Never	1565 (51.77)	3488 (95.85)	1751.88	< 0.001
Former	652 (21.57)	74 (2.03)		
Current	806 (26.66)	77 (2.12)		
Lifestyle behaviours				
Sedentary				
< 3 h/day	1191 (38.42)	1477 (40.59)	1.82	0.610
3–5 h/day	1182 (39.13)	1404 (38.58)		
> 5 h/day	648 (21.45)	758 (20.83)		
Physical activity				
< 3 times/week	978 (32.35)	987 (27.11)	21.85	< 0.001
≥ 3 times/week	2045 (67.65)	2654 (72.89)		
Healthy diet				
Yes	2981 (98.64)	3574 (98.21)	1.95	0.160
No	41 (1.36)	65 (1.79)		
Sleep behaviours				
Sleep disturbance				
0 times/week	1710 (56.57)	1700 (46.73)	72.43	< 0.001
1–2 times/week	953 (31.52)	1314 (36.12)		
3–4 times/week	239 (7.91)	404 (11.11)		
5–7 times/week	121 (4.00)	220 (6.05)		
Sleep duration				
< 6 h/day	730 (24.15)	1137 (31.24)	42.83	< 0.001
6–8 h/day	2080 (68.81)	2297 (63.10)		
> 8 h/day	213 (7.05)	206 (5.66)		
Diseases number				
0	1075 (35.58)	1164 (32.00)	18.24	0.001
1	1518 (50.25)	1857 (51.04)		
2	363 (12.02)	533 (14.65)		
≥ 3	65 (2.15)	84 (2.31)		

Table 1. Baseline characteristics in men and women.

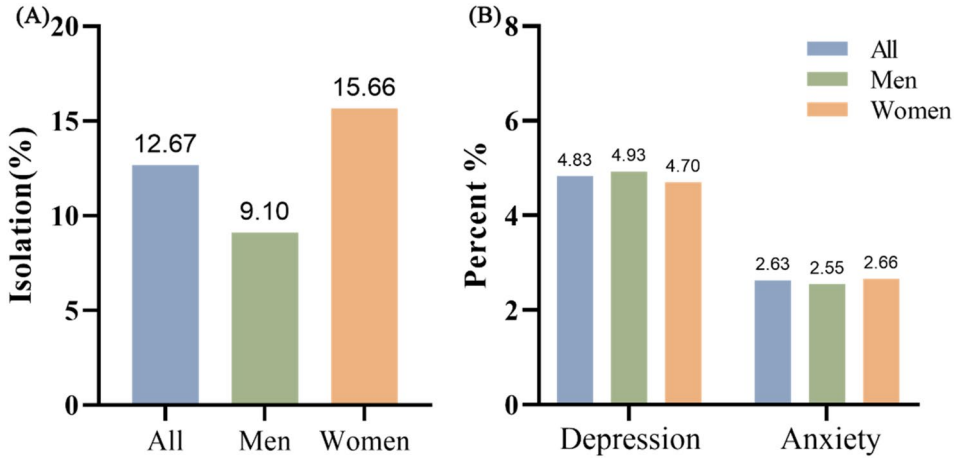

Figure 1. The percentage of isolation (**A**), anxiety and depression (**B**) in the gender group among over 65 years old Chinese.

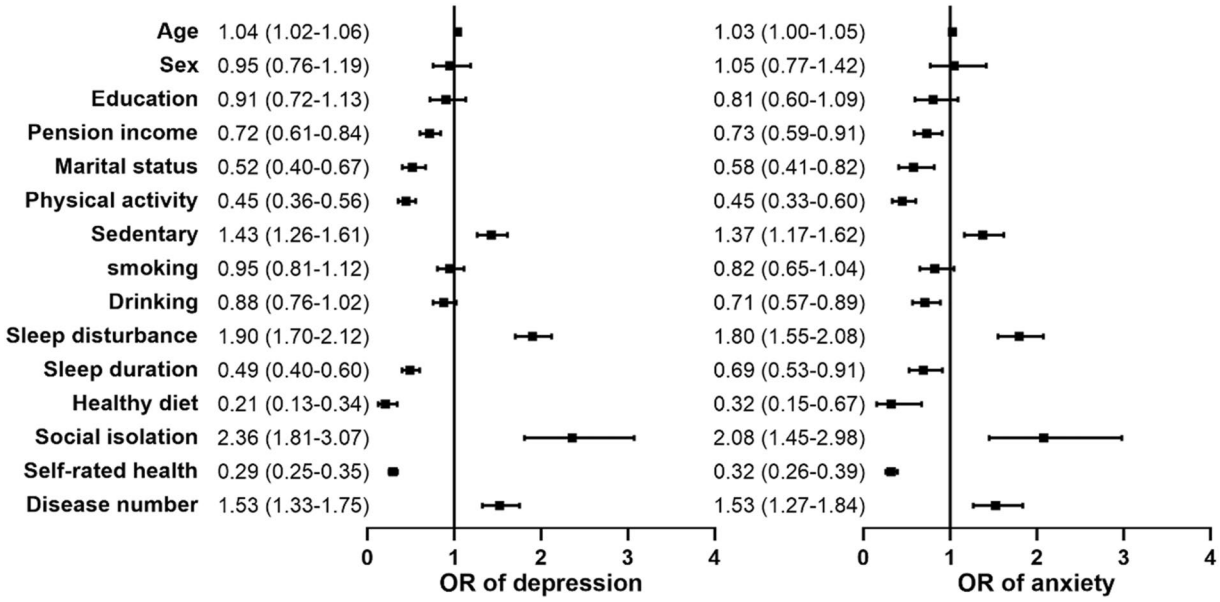

Figure 2. The association factors of anxiety and depression by univariate logistic regression analysis.

	Depression		Anxiety	
	AOR[a] (95%CI)	*P* value	AOR[a] (95%CI)	*P* value
All	1.77 (1.25–2.51)	0.001	1.66 (1.05–2.63)	0.029
Men	1.42 (0.77–2.58)	0.258	0.98 (0.42–2.26)	0.963
Women	1.99 (1.28–3.09)	0.002	1.92 (1.08–3.42)	0.026

Table 2. The multivariate adjusted association between social isolation and anxiety and depression. [a]Multivariate adjusting for age, sex, education, pension income, marital status, self-rated health, sedentary, physical activity, drinking, smoking, sleep disturbance, sleep duration, healthy diet, and disease number.

provision of companionship or care for them, (3) offering opportunities for them to engage in social interaction (e.g., community activities), and (4) changing their cognition about social isolation and social support[38,39]. Social isolation is the objective absence of social relations, meanwhile, loneliness is the subjective experience of the lack of social relations[35]. Although social isolation and loneliness are not exactly equivalent, potential interventions for eliminating social isolation among older adults might partly mitigate their loneliness, and further alleviate mental unhealth[40].

Dependent variables	Isolation exposure	No	Percentage	AOR[a] (95%CI)	PAR%
Depression	No	244	0.755	1	
	Yes	79	0.245	1.77 (1.25–2.51)	10.66
Anxiety	No	136	0.773	1	
	Yes	40	0.227	1.66 (1.05–2.63)	9.03

Table 3. The multivariate adjusted population attributable risk of social isolation for depression and anxiety. [a]Multivariate adjusting variables as Table 2. PAR%: Adjusted population attributable risk percentage.

Sex	Dependent variables	Isolation exposure	No	Percentage	AOR[a] (95%CI)	PAR%
Men	Depression	No	125	0.822	1	
		Yes	27	0.178	1.42 (0.77–2.58)	5.26
	Anxiety	No	65	0.823	1	
		Yes	14	0.177	0.98 (0.42–2.26)	0
Women	Depression	No	119	0.696	1	
		Yes	52	0.304	1.99 (1.28–3.09)	15.12
	Anxiety	No	71	0.732	1	
		Yes	26	0.268	1.92 (1.08–3.42)	12.84

Table 4. The multivariate adjusted population attributable risk of social isolation for depression and anxiety by sex. [a]Multivariate adjusting variables as Table 2. PAR%: Adjusted population attributable risk percentage.

It is evident that elderly women are more likely to experience social isolation. In 2019, the average life expectancy in China was 77.7 years, with a gender disparity of 6.2 years favoring women. This gap is projected to widen to 7.0 years by 2035[41]. Consequently, elderly women are more likely to experience widowhood and live alone, thereby heightening their risk of social isolation. Furthermore, the association and PAR% of social isolation with depression and anxiety were found to be significant exclusively among elderly women. Gender differences in response to social isolation may be partially attributable to sexually dimorphic reactions. In male mice subjected to social isolation, increased dopamine release was observed during subsequent social interactions, leading to enhanced pleasurable sensations. This, in turn, fostered a heightened desire for social engagement[42]. The phenomenon was not observed in female mice. Consequently, this can explain the reason for the impact of social isolation appeared to be mitigated in males.

Strengths and limitations

The strengths of the current study include a large sample size and the use of validated scales to measure social isolation, depression, and anxiety. In addition, the PAR% of social isolation in depression and anxiety was first explored. However, this study has several limitations. First, all of the participants were recruited from communities in Ningbo, and therefore the conclusions may not be generalizable to other regions in China. Second, this was a cross-sectional study, and therefore causal associations cannot be assumed. Longitudinal studies of the relationships of social isolation with depression and anxiety will be needed to infer causality. In addition. although we adjusted for as many confounders as possible, this was an observational study, and therefore we cannot rule out the possibility of residual confounders. Third, all of the data included in the analyses were based on self-reports, suggesting the possibility of reporting bias. The scales for depression and anxiety were based on self-reports rather than clinical assessments and diagnoses, which may also contribute to different results. Fourth, some of the participants had poor health status (such as having difficulty in reading or writing) and required assistance from the interviewers to complete their questionnaires, which may have increased the risk of social desirability bias.

Conclusions

In conclusion, this study highlights the associations of social isolation with increased risks of depression and anxiety among older adults, especially among females. The findings from this study suggest the importance of identifying older adults who are at risk of social isolation, especially among females. Effective and feasible interventions are needed to eliminate social isolation and pay attention to mental health in this population.

Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at Affiliated Kangning Hospital of Ningbo University.

Received: 25 January 2024; Accepted: 12 August 2024
Published online: 17 August 2024

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Acknowledgements
The authors express their gratitude to the doctors from the office of Mental health and the community health service center who were involved in the data collection.

Author contributions
L.L. conceived and designed the study, performed the statistical analysis, and drafted the article. K.P., J.L., M.J., and Y.G. collected the data. Hongying Yang and Guolin Bian contributed to the study design and the article's critical revision. All authors revised the article and approved the final version.

Funding
This work was supported by the Ningbo Medical & Health Leading Academic Discipline Project (2022-F28), Ningbo Top Medical and Health Research Program (2022030410) and Ningbo Medical and Health Brand Discipline (PPXK2018-08).

Competing interests
The authors declare no competing interests.

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